

A Real-time Vehicle Detection Pipeline with Data-centric Enhancements and Multi-stage DETR Distillation

Huy Nguyen Minh Nhat Hieu Pham Dinh Trung Khang Le Minh Cuong Nguyen Tuan
Vietnam–Germany University (VGU)
Ho Chi Minh City, Vietnam

{10423045, 104240027, 104240055}@student.vgu.edu.vn, cuong.nt2@vgu.edu.vn

Abstract

Real-time vehicle detection often requires trading off accuracy for speed. To validate a solution that excels on both fronts, we adopt fisheye imagery, a domain where extreme radial distortion and scale variation defeat standard detectors, as a rigorous testbed. Our pipeline comprises three key stages: (1) Multi-stage DETR Distillation, a four-phase knowledge transfer leveraging KD-DETR’s fixed “distillation queries,” [18] with separate head-level and feature-level distillation stages to avoid gradient conflicts and ensure progressive learning; (2) Data-centric Enhancements, creating a diverse training pool via Co-DETR[21] pseudo-labeling, CycleGAN-Turbo[10] day-to-night style transfer, and object-level flash/blur augmentations; and (3) Adaptive Sample Mining, which dynamically upsamples complex examples to sharpen the model’s focus. Evaluated on the challenging FishEye8K benchmark, our approach delivers strong accuracy while maintaining real-time FPS on constrained devices. By “ignoring” fisheye distortions and treating them as a domain-agnostic stress test, we demonstrate that this data-centric, multi-stage distillation framework generalizes seamlessly to standard vehicle and broader object detection tasks, offering a unified solution for high-precision, real-time vision systems.

1. Introduction

Building robust vehicle detection systems that operate reliably across diverse visual domains such as urban scenes, aerial views, and wide-angle imagery is a key challenge for real-world deployment. Fisheye imagery represents an extreme case among these domains due to its strong radial distortion and severe scale variation. These distortions, especially near image boundaries, stretch or compress objects into irregular shapes, often shrinking them to just a few pixels. As a result, conventional detectors trained on undis-

torted images struggle in such settings, leading to sharp performance degradation. In this work, we focus on fisheye as a challenging testbed to evaluate the generalization and adaptability of our real-time detection pipeline.

Most prior work in fisheye object detection has primarily focused on improving accuracy, often through complex architectures or ensembling multiple models. While these methods can yield strong performance on benchmark datasets, they frequently come at the cost of high computational overhead, making them unsuitable for deployment on edge devices with limited resources. This creates a practical gap between research and real-world applications, where accuracy and low latency are equally critical.

We address this gap with a data-centric and distillation pipeline that leverages the powerful concept of distillation points from KD-DETR[18] and organizes knowledge transfer into multiple distinct stages to avoid gradient conflicts and maximize learning efficiency. Our key contributions are:

- **Multi-stage DETR Distillation:** We fully exploit KD-DETR’s fixed “distillation queries” [18] to anchor teacher-student alignment. Instead of a single distillation pass, we split knowledge transfer into four sequential stages: head-level logits, encoder-feature alignment, joint refinement, and final fine-tuning, ensuring stable gradients and progressive absorption of teacher expertise.
- **Data-centric Enhancements:** We enrich FishEye8K[5] with VisDrone[3] and LOAF[19] datasets, use Co-DETR[21] to fill in vehicle pseudo-labels on LOAF. Finally, we apply CycleGAN-Turbo[10] for day-to-night transfer, adding flash and motion-blur effects to expand and diversify our training data without changing any network architecture.
- **Adaptive Sample Mining:** Inspired by Online Hard Example Mining[15], we dynamically up-sample training samples with low F1-scores, forcing the student to focus on its weakest cases and further tune its capacity to challenging scenarios.

By integrating these stages carefully, disentangled to prevent conflicting updates, we achieve high accuracy on FishEye8K[5] while preserving real-time FPS. This stage-wise pipeline shows that carefully crafted data strategies and multi-stage distillation can deliver superior results without complex architectural modifications. Although our pipeline focuses on fisheye distortion, the proposed data-centric, multi-stage distillation pipeline broadly applies to general vehicle detection. Its sensor-agnostic design makes it suitable for various real-time, high-accuracy object detection tasks beyond the fisheye domain.

2. Related Work

2.1. Real-time Detection with Transformers

Real-time object detection has advanced from fast CNN-based models such as YOLO[13] and SSD[8] to Transformer-based approaches such as DETR[1], which remove anchors and NMS for end-to-end prediction. However, DETRs often suffer from high latency and slow convergence. D-FINE[11] addresses this by refining the bounding box regression through probability distributions and introducing a lightweight self-distillation mechanism across decoder layers. These designs improve localization accuracy while preserving real-time performance.

2.2. Fisheye Object Detection

Fisheye cameras offer wide-angle coverage, ideal for surveillance and traffic monitoring, but introduce substantial geometric distortions that challenge conventional detectors. These distortions affect object shape, size, and location, especially near image edges, requiring models to be aware of the distortion or trained with specialized data. As a result, fisheye object detection remains a challenging task that requires robust and adaptive detection frameworks.

2.3. Knowledge Distillation for Transformer-Based Detectors

Knowledge distillation (KD)[6] enables compact student models to learn from larger teacher models by mimicking outputs or intermediate features. Although effective in CNN-based detectors, applying KD to Transformer-based detectors like DETR[1] is challenging due to the unstable one-to-one bipartite matching between queries and objects.

DETRDistill[2] extends classic KD by matching queries via the Hungarian algorithm, but suffers from inconsistent teacher-student correspondences[18]. KD-DETR[18] addresses this by introducing fixed "distillation queries" shared between teacher and student, ensuring stable supervision. It further enhances transfer by combining random and teacher-derived queries and applies a foreground-weighted loss to emphasize object-relevant regions.

2.4. AI City Challenge Approaches

In recent AI City Challenge[17], many top-performing solutions combined diverse strategies to enhance detection under real-world constraints. Duong et al.[4] adopted robust data augmentation, external datasets such as VisDrone[3], and ensemble techniques to improve robustness. Pham et al.[12] leveraged CycleGAN[20] to synthesize night-time images, addressing illumination gaps. Co-DETR[21], a strong Transformer-based detector, was widely adopted across teams due to its effectiveness, despite not being optimized for real-time inference.

3. Proposed Method

3.1. Datasets and Models Selection

3.1.1. Dataset

In addition to the official FishEye8K dataset[5], we incorporate VisDrone[3] and LOAF[19] datasets to enhance robustness and diversity. VisDrone is a widely used aerial object detection dataset adopted by several top-ranking teams in the previous AI City Challenge[17]. It complements FishEye8K by introducing varied viewing angles and richer urban scenes. As LOAF is focused on pedestrians, incorporating it helps the model better learn and generalize pedestrian characteristics in the fisheye setting of FishEye8K.

3.1.2. Model Selection

We explore multiple Transformer-based detectors and adopt different models for complementary roles. Co-DETR[21] was selected for generating pseudo-labels, given its strong performance in previous competitions and its use by several top teams [4][12]. However, its high computational cost makes it unsuitable for real-time deployment in our pipeline.

Our main detector is D-FINE[11], which redefines bounding-box regression as fine-grained distribution refinement and introduces a Global Optimal Localization Self-Distillation (GO-LSD). D-FINE[11] delivers a favorable balance between speed and accuracy, making it well-suited for real-time fisheye object detection without additional inference overhead.

3.2. Data-centric Enhancements

In this section we introduce our data processing methods to create Data Pool as visualized in Figure 1.

Pseudo label: Manual annotation errors, such as inconsistent bounding box extents across similar scenes, can substantially degrade lightweight detectors. To remedy this, we first train a state-of-the-art Co-DETR model [21] on VisDrone [3] and Fisheye8K [5] using a 3-fold cross-validation split over distinct camera clusters, combined with strong geometric and photometric enhancements, test time enhancement, and weighted box fusion (WBF) [16] that assembles

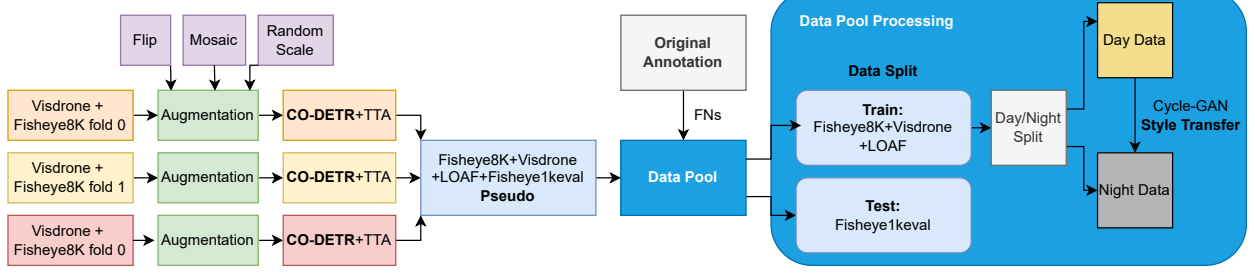


Figure 1. Our Data-centric Framework

to produce high-quality pseudo-labels. We then compare each pseudo-box with the original manual annotation via an IoU threshold (e.g., 0.5) and reinsert any false negatives in the training set. This semi-supervised refinement not only cleanses the original annotations but also augments the LOAF dataset [19] (originally annotated only for pedestrians), by recovering vehicle labels.

Day and Night Divide and Conquer: Based on the Divide and Conquer and Mixture of Experts, we split our dataset into day and night parts. We will train explicitly 2 models for each part, then activate both of them in the inference, choose 1 to run each image by the time or image color distribution. But because of the imbalance of day and night images in the challenge dataset, we used pretrained *day-to-night* CycleGAN-Turbo[10] as a style transfer module to increase night images from the day subset.

3.3. Adaptive Sample Mining

Beyond conventional data augmentations, we introduce *Adaptive Sample Mining*, a dynamic training strategy inspired by Online Hard Example Mining [?] but simplified for ease of integration. After an initial burn-in period of a few epochs, by which time the model’s loss has stabilized, we evaluate the F1-score of each training sample against its refined labels. Samples with F1-scores below a predefined threshold are then duplicated (or up-sampled) in the subsequent epoch’s training set, thereby exponentially increasing the model’s exposure to difficult cases. This adaptive re-sampling drives the student to focus learning capacity on its weakest examples. While effective, this procedure increases training time; to balance accuracy gains against computational cost, we apply Adaptive Sample Mining only after convergence of the base loss and cap the maximum replication factor per sample.

3.4. Multi-stage DETR Distillation

We build upon KD-DETR’s insight that a fixed, shared set of *distillation points* $\tilde{q}$ can act as reliable anchors for transferring both logit- and feature-level knowledge between teacher and student in transformer detectors[18]. Our framework unfolds over four carefully designed stages,

each focusing on a different aspect of learning while always ensuring the student decodes its own representations.

First, we construct $\tilde{q} \in \mathbb{R}^{M \times d}$ by concatenating M_g ‘general’ queries - randomly initialized embeddings re-sampled every iteration to probe the full encoder feature map with M_s ‘specific’ queries drawn from the teacher’s own learned object queries. As in KD-DETR[18], each point $\tilde{q}_i$ is weighted by the teacher’s maximum class probability $w_i = \max_c p_T(y_c | \tilde{q}_i)$, so that distillation focuses on the most informative, foreground-rich locations.

Stage 1: Initialization Both teacher and student are pre-trained end-to-end on the dataset with 3 schedules (12 epochs per schedule).

Stage 2: Logits Exploitation We freeze the teacher together with the student’s backbone and encoder, unfreezing only the student’s decoder heads. Both models decode the student’s feature map $F_S(I)$ with $\tilde{q}$ to produce (c_T, b_T) and (c_S, b_S) , and the student minimizes

$$\mathcal{L}_{\text{distill}} = \sum_{i: \max c_T^i > \tau} w_i \left[\alpha_{cls} L_{\text{KL}}(c_T^i \| c_S^i) + \alpha_1 \|b_S^i - b_T^i\|_1 + \alpha_{\text{GIoU}} \text{GIoU}(b_S^i, b_T^i) \right], \quad (1)$$

while still incurring $\mathcal{L}_{\text{det}}$ on its original queries. By mimicking the teacher’s class and box predictions at each distillation point, the student head precisely absorbs the teacher’s output behavior.

Stage 3: Feature-level Exploitation All decoder heads (teacher and student) are frozen, and we unfreeze only the student’s backbone and encoder. The student again incurs the standard detection loss $\mathcal{L}_{\text{det}}$ on its original queries, while feature alignment is enforced via a non-parametric attention-pooling operation at each $\tilde{q}_i$. Flattening the encoder output $F \in \mathbb{R}^{C \times H \times W}$ into key/value matrices $K = V \in \mathbb{R}^{HW \times d}$, each query yields

$$\alpha_i = \text{softmax}\left(\frac{\tilde{q}_i K^\top}{\sqrt{d}}\right), \quad z^i = \alpha_i V,$$

and we minimize the feature-level loss

$$\mathcal{L}_{\text{feat}} = \frac{1}{M} \sum_{i=1}^M \|z_S^i - z_T^i\|_2^2. \quad (2)$$

This encourages the student encoder to reproduce the teacher’s internal representations at exactly the same distillation points.

Stage 4: Exploration Finally, we unfreeze all student parameters and perform a full KD-DETR architecture [18] distillation pass, omitting feature distillation. The loss is given by

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_h \mathcal{L}_{\text{distill}}. \quad (3)$$

This last phase reconciles detection accuracy with faithful head-level matching to the teacher, consolidating all previously distilled knowledge into a single, cohesive update.

Through this sequence-supervised initialization, targeted head-level distillation, encoder-focused feature training, and joint refinement, we leverage KD-DETR’s distillation points most effectively, achieving stable, high-fidelity knowledge transfer across diverse DETR variants.

4. Experiment Results and Discussion

4.1. Evaluation Metrics

We evaluated our system using three metrics: F1-score, FPS, and their harmonic mean.

F1-score balances precision and recall, reflecting the model’s ability to detect objects accurately while minimizing false detections and misses.

FPS (Frames Per Second) measures inference speed, indicating real-time capability.

Harmonic Mean of F1 and FPS, denoted as $HM_{F1,FPS}$, is the primary ranking metric:

$$HM_{F1,FPS} = \frac{2 \cdot F1 \cdot FPS}{F1 + FPS}.$$

This metric penalizes models that excel in only one aspect, promoting practical, real-time solutions that are accurate and fast.

4.2. Dataset and Model

4.2.1. Datasets

We follow the top-ranked team on AI City Challenge 2024 - Track 4[17][4] by splitting FishEye8K[5] into three folds for cross-validation. Fold 0 uses the organizers’ default split; folds 1 and 2 are partitioned by camera IDs to maintain a 70:30 train-val ratio of object counts, with no video overlap between folds.

To augment robustness, we incorporate two external datasets:

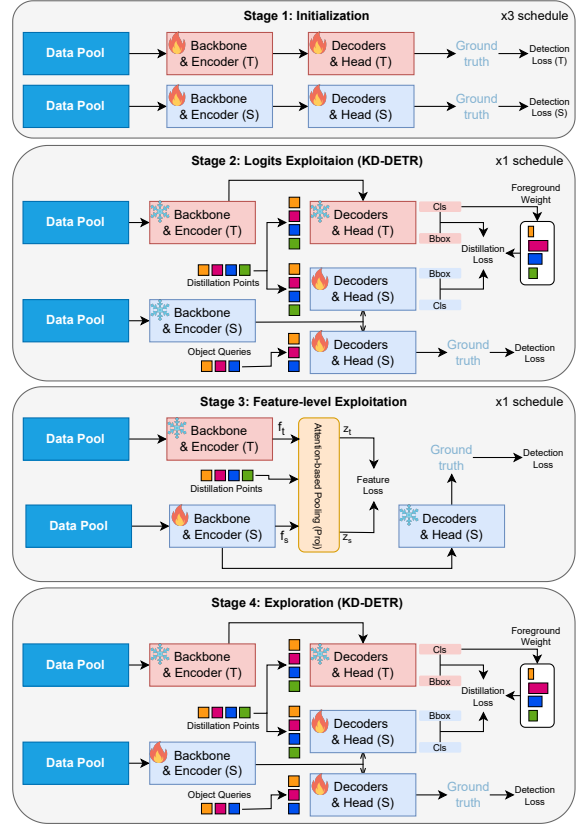


Figure 2. Our Multi-stage DETR Distillation Architecture

- **VisDrone** [3]: Chosen because last year’s winning solution demonstrated significant gains when pre-trained on the VisDrone dataset.
- **LOAF** [19]: A large-scale overhead fisheye benchmark focused on pedestrians. We relabel its ‘person’ annotations as ‘pedestrian’ and retain only a balanced subset of 20K images containing vehicles (car, bus, truck, bike). The remaining object classes are pseudo-labeled using our Co-DETR model[21], which we trained on section ??

4.2.2. Implementation Details

All experiments are conducted on a single NVIDIA RTX 4090 GPU, with input resolution fixed at 1280×1280 . Every model is initialized from weights pre-trained on COCO[7] and Objects365[14]. Table ?? summarizes the training configurations (learning rate, batch size, epochs) and evaluation results (F1, FPS) of our main models.

Model	Data	F1-score	FPS
Co-DETR	VisDrone + FishEye8K (fold 0)	0.51	2
D-FINE-X	VisDrone + FishEye8K + LOAF-20K	0.46	47
D-FINE-X	VisDrone + FishEye8K	0.38	47
D-FINE-L	VisDrone + FishEye8K	0.33	79
YOLOv11-X	VisDrone + FishEye8K (fold 0)	0.35	38
YOLOv11-L	VisDrone + FishEye8K (fold 0)	0.31	72

Table 1. Performance of several models.

4.3. Quantitative Results

To assess the impact of our data-centric enhancements, adaptive sample mining, and multi-stage DETR distillation, we conducted a series of experiments and ablation studies.

Pseudo-label: To evaluate the feasibility of the pseudo-labeling method introduced earlier, we followed the pipeline in Figure 1 up to the Data Pool creation step, but applied it only to the Visdrone + Fisheye8k_{train} set. Table 2 presents a comparative analysis of various models trained with and without pseudo labels. The results indicate that most models benefit from pseudo-labeling, with the smallest architectures showing the largest relative gains, demonstrating that relabeling the entire dataset effectively reduces noise and helps compact models focus on the most salient object features.

Model	Pseudo data	F1-score	FPS
Co-DETR	✓	0.4628	2
Co-DETR	—	0.4613	2
D-Fine-X	✓	0.3925	47
D-Fine-X	—	0.3888	47
D-Fine-L	✓	0.3477	79
D-Fine-L	—	0.3361	79
D-Fine-M	✓	0.3350	145
D-Fine-M	—	0.3215	145
D-Fine-S	✓	0.2925	221
D-Fine-S	—	0.2866	221
D-Fine-N	✓	0.2577	302
D-Fine-N	—	0.2231	302

Table 2. Impact of pseudo-labeling on performance.

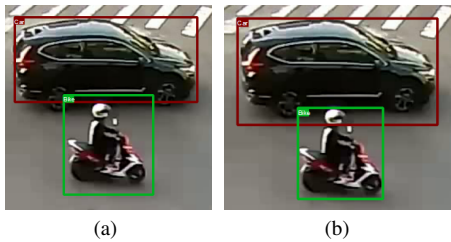


Figure 3. **Example of pseudo label using Co-DETR** (a) Original image. (b) Pseudo labeling image

Day–Night Divide and Conquer To boost night-scene performance while retaining real-time speed, we package two steps into a single Day–Night Module: **(i) Style Transfer & Augmentation.** Daytime fisheye frames are converted to pseudo-night images with CycleGAN-Turbo; large objects are probabilistically enhanced by grayscale, flash, and motion-blur effects (Table 3). **(ii) Lightweight Routing.** At inference, each frame is dispatched to the corresponding expert via a filename cue (`_N/_E_` vs. `_M/_A_`) or a grayscale fallback when no cue is present.

Dataset	p_{flash}	A_{flash}	p_{blur}	A_{blur}
Fisheye8K	0.5	1500	0.75	3000
Fisheye1K	0.4	2000	0.75	3000
Visdrone dataset	0.4	1500	0.75	3000
LOAF-20K	0.2	2000	0.75	3000

Table 3

Figure 4 illustrates the style-transfer pipeline, and Table 4 confirms that the module lifts F1 from 0.67 to 0.69 without compromising speed as the test set contains 10 percent night images in total, which mean it has a lot of potential in the balancing day-night test set.

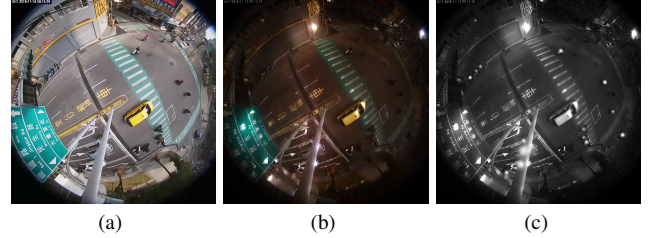


Figure 4. **Example of nighttime data generation using style transfer.** (a) Original daytime images. (b) Generated nighttime images. (c) Generated nighttime images with grayscale, flash, and motion blur augmentation.

Model	Classification method	F1-score	FPS
D-Fine-X		0.67	47
D-Fine-X _{day-night}	file name	0.69	47
D-Fine-X _{day-night}	grayscale	0.69	45

Table 4. Day–Night Divide and Conquer Performance

After finalizing the data-centric enhancement pipeline visualized in Figure 1, we conduct an in-depth evaluation of our **Multi-stage DETR Distillation** method. We first train D-FINE X on our Data Pool, achieving an F_1 score of 0.67; this model then serves as our largest teacher. Our goal is to help smaller student variants approach D-FINE X’s performance.

First, we experiment with D-FINE L as the student. Table 5 details the effectiveness of each distillation stage compared to the original KD-DETR (which only uses stages 1 and 4). Figure 5 shows the attention maps of both teacher and student after distillation. Table 6 summarizes the performance gains of all D-FINE variants under our multi-stage DETR distillation. Knowledge is distilled successively from the largest model to progressively smaller ones ($X \rightarrow L \rightarrow M$). We also validate our approach on the RT-DETRv2[9] backbone and observe similar gains, demonstrating that our method is truly *architecture-agnostic*, i.e., it can be applied to any DETR variant without modifying its original architecture.

Model	Stage 1	Stage 2	Stage 3	Stage 4	F1-score
D-Fine-X (T)	✓	—	—	—	0.6700
D-Fine-L (S)	✓	—	—	—	0.6412
D-Fine-L (S)	✓	✓	—	—	0.6528
D-Fine-L (S)	✓	✓	✓	—	0.6571
D-FINE-L (S)	✓	✓	✓	✓	0.6636
D-FINE-L (S)	✓	—	—	✓	0.6576
RT-DETRv2-X (T)	✓	—	—	—	0.6320
RT-DETRv2-L (S)	✓	—	—	—	0.5915
RT-DETRv2-L (S)	✓	✓	—	—	0.6018
RT-DETRv2-L (S)	✓	✓	✓	—	0.6043
RT-DETRv2-L (S)	✓	✓	✓	✓	0.6216
RT-DETRv2-L (S)	✓	—	—	✓	0.6207

Table 5. Impact of multi-stage distillation on performance.

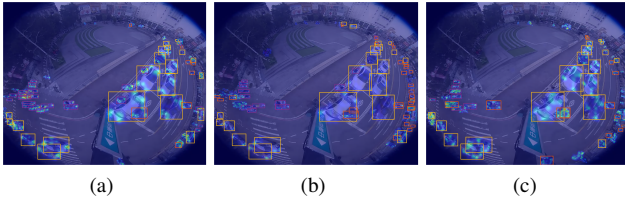


Figure 5. **Attention view of Multi-stage DETR Distillation** (a) Teacher. (b) Student before. (c) Student after.

For the challenge, we submitted all D-FINE variants, each enhanced by our data-centric pipeline and multi-stage distillation, to the general leaderboard. Table 7 reports their performance on the original full test set. Since these results were not sufficiently competitive, we selected the D-FINE M model (F1-score = 0.6230, 145 FPS) for final tuning using the **Adaptive Sample Mining** strategy described in Section 3.3. This process yielded an improved F_1 score of 0.6318 for D-FINE M, as presented in Table 8.

5. Conclusion

We have introduced a cohesive, three-part framework that significantly improves DETR-based object detectors without any architectural modifications. First, our data-centric

Model	F1-score	FPS
D-Fine-X	0.670	47
D-Fine-L	0.6471	79
D-Fine-L + distill	0.6636 (+0.02)	79
D-Fine-M	0.6231	145
D-Fine-M + distill	0.6412 (+0.02)	145
D-Fine-S	0.5886	221
D-Fine-S + distill	0.605 (+0.01)	221
D-Fine-N	0.5514	302
D-Fine-N + distill	0.5871 (+0.03)	302

Table 6. Improvement from multi-stage distillation across model scales.

Model	AP _{0.5:0.95}	AP _{0.5}	F1-score
D-Fine-X	0.5757	0.7878	0.6324
D-Fine-L	0.5603	0.7771	0.6259
D-Fine-M	0.5410	0.7743	0.6230
D-Fine-S	0.5287	0.7294	0.5851
D-Fine-N	0.4941	0.6842	0.5597

Table 7. Detection performance across model scales on original test set.

Rank	Team Name	Score	Rank	Team Name	Score
1	UIT-OpenCubee	0.6493	8	Smart Lab	0.6366
2	UT.T1	0.6413	9	MIZSU	0.6360
3	UT.KAO	0.6413	10	SmartVision	0.6342
4	Zacian	0.6405	11	UT.Wu	0.6330
5	SKKU-AutoLab	0.6397	12	VGU-VLInsight	0.6318
6	UT.Liu	0.6393	13	Tyche	0.6302
7	ttsc	0.6381	14	xiilab	0.6268

Table 8. Final test set evaluation of AI City Track 4 Challenge

enhancement pipeline generates diverse, realistic nighttime variants from daytime images, boosting domain robustness. Second, our multi-stage DETR distillation transfers knowledge progressively from the largest teacher to smaller students, isolating updates to each module and avoiding gradient conflicts. Third, Adaptive Sample Mining fine-tunes the student by prioritizing hard and under-represented examples, yielding additional F1-score gains. Across D-FINE L, M, and RT-DETR, we observe consistent improvements and achieve an F1-score of 0.6318 at 145 FPS for D-FINE M on the challenge leaderboard. These results demonstrate both the practical effectiveness and the broad applicability of our approach. Future work will explore self-supervised pretraining and dynamic distillation schedules to further enhance robustness and efficiency in real-world deployments.

References

- [1] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. *CoRR*, abs/2005.12872, 2020. 2
- [2] Jiahao Chang, Shuo Wang, Haiming Xu, Zehui Chen, Chenhongyi Yang, and Feng Zhao. Detrdistill: A universal knowledge distillation framework for detr-families, 2023. 2
- [3] Dawei Du, Pengfei Zhu, Longyin Wen, Xiao Bian, Haibin Lin, Qinghua Hu, Tao Peng, Jiayu Zheng, Xinyao Wang, Yue Zhang, Liefeng Bo, Hailin Shi, Rui Zhu, Aashish Kumar, Aijin Li, Almaz Zinollayev, Anuar Askergaliyev, Arne Schumann, Binjie Mao, Byeongwon Lee, Chang Liu, Changrui Chen, Chunhong Pan, Chunlei Huo, Da Yu, DeChun Cong, Dening Zeng, Dheeraj Reddy Pailla, Di Li, Dong Wang, Donghyeon Cho, Dongyu Zhang, Furui Bai, George Jose, Guangyu Gao, Guizhong Liu, Haitao Xiong, Hao Qi, Haoran Wang, Heqian Qiu, HongLiang Li, Huchuan Lu, Ildoo Kim, Jaekyum Kim, Jane Shen, Jihoon Lee, Jing Ge, Jingjing Xu, Jingkai Zhou, Jonas Meier, Jun Won Choi, Junhao Hu, Junyi Zhang, Junying Huang, Kaiqi Huang, Keyang Wang, Lars Sommer, Lei Jin, and Lei Zhang. Visdrone-det2019: The vision meets drone object detection in image challenge results. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, 2019. 1, 2, 4
- [4] Viet Hung Duong, Duc Quyen Nguyen, Thien Van Luong, Huan Vu, and Tien Cuong Nguyen. Robust data augmentation and ensemble method for object detection in fisheye camera images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 7017–7026, 2024. 2, 4
- [5] Munkhjargal Gochoo, Munkh-Erdene Otgonbold, Erkhembayar Ganbold, Jun-Wei Hsieh, Ming-Ching Chang, Ping-Yang Chen, Byambaa Dorj, Hamad Al Jassmi, Ganzorig Batnasan, Fady Alnajar, Mohammed Abduljabbar, and Fang-Pang Lin. Fisheye8k: A benchmark and dataset for fisheye camera object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 5305–5313, 2023. 1, 2, 4
- [6] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015. 2
- [7] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context, 2015. 4
- [8] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott E. Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: single shot multibox detector. *CoRR*, abs/1512.02325, 2015. 2
- [9] Wenyu Lv, Yian Zhao, Qinyao Chang, Kui Huang, Guanzhong Wang, and Yi Liu. Rt-detr2: Improved baseline with bag-of-freebies for real-time detection transformer, 2024. 6
- [10] Gaurav Parmar, Taesung Park, Srinivasa Narasimhan, and Jun-Yan Zhu. One-step image translation with text-to-image models, 2024. 1, 3
- [11] Yansong Peng, Hebei Li, Peixi Wu, Yueyi Zhang, Xiaoyan Sun, and Feng Wu. D-fine: Redefine regression task in detr as fine-grained distribution refinement, 2024. 2
- [12] Long Hoang Pham, Quoc Pham-Nam Ho, Duong Nguyen-Ngoc Tran, Tai Huu-Phuong Tran, Huy-Hung Nguyen, Duong Khac Vu, Chi Dai Tran, Ngoc Doan-Minh Huynh, Hyung-Min Jeon, Hyung-Joon Jeon, and Jae Wook Jeon. Improving object detection to fisheye cameras with open-vocabulary pseudo-label approach. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 7100–7107, 2024. 2
- [13] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection, 2016. 2
- [14] Shuai Shao, Zeming Li, Tianyuan Zhang, Chao Peng, Gang Yu, Jing Li, Xiangyu Zhang, and Jian Sun. Objects365: A large-scale, high-quality dataset for object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8425–8434, 2019. 4
- [15] Abhinav Shrivastava, Abhinav Gupta, and Ross Girshick. Training region-based object detectors with online hard example mining, 2016. 1
- [16] Roman Solovyev, Weimin Wang, and Tatiana Gabruseva. Weighted boxes fusion: Ensembling boxes from different object detection models. *Image and Vision Computing*, 107: 104117, 2021. 2
- [17] Shuo Wang, David C. Anastasiu, Zheng Tang, Ming-Ching Chang, Yue Yao, Liang Zheng, Mohammed Shaiqur Rahman, Meenakshi S. Arya, Anuj Sharma, Pranamesh Chakraborty, Sanjita Prajapati, Quan Kong, Norimasa Kobori, Munkhjargal Gochoo, Munkh-Erdene Otgonbold, Ganzorig Batnasan, Fady Alnajar, Ping-Yang Chen, Jun-Wei Hsieh, Xunlei Wu, Sameer Satish Pusegaonkar, Yizhou Wang, Sujit Biswas, and Rama Chellappa. The 8th AI City Challenge. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2024. 2, 4
- [18] Yu Wang, Xin Li, Shengzhao Weng, Gang Zhang, Haixiao Yue, Haocheng Feng, Junyu Han, and Errui Ding. Kd-detr: Knowledge distillation for detection transformer with consistent distillation points sampling, 2025. 1, 2, 3, 4
- [19] Lu Yang, Liulei Li, Xueshi Xin, Yifan Sun, Qing Song, and Wenguan Wang. Large-scale person detection and localization using overhead fisheye cameras, 2023. 1, 2, 3, 4
- [20] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks, 2020. 2
- [21] Zhuofan Zong, Guanglu Song, and Yu Liu. Detr with collaborative hybrid assignments training, 2023. 1, 2, 4